NATIONAL SCHOOL OF BUSINESS MANAGEMENT
Year 01 Semester 02 Examination
SE101.3- Object Oriented Programming with Java
Model Paper

Instructions to Candidates

1) **Answer All Questions (Part A & B)**
2) Illustrate answers with sketches where required.
3) The time allocated for the examination is **three (03) hours.**
4) If a page or a part of this question paper is not printed, please inform the Supervisor immediately.
5) Write your index number on all pages of the answer script.
6) Mark the correct answer on the given answer grid.

1. What is the default value of an uninitialized boolean instance variable in Java?
    a. true
    b. false
    c. 0
    d. null
2. Which of the following variable types is shared across all objects of a class?
    a. Local variable
    b. Instance variable
    c. Static variables
    d. Reference variables
3. Which of the following is not an OOP concept in Java?
    a. Polymorphism
    b. Inheritance
    c. Compilation
    d. Encapsulation
4. Which component of Java executes bytecode?
    a. Compiler
    b. Interpreter
    c. JVM (Java Virtual Machine)
    d. JDK (Java Development Kit)
5. What is the main purpose of the main() method in a Java program?
    a. To define global variables
    b. To serve as the entry point of the application
    c. To declare classes
    d. To initialize attributes
6. Which extension is used for compiled Java bytecode files?
    a. .java
    b. .class
    c. .jar
    d. .exe
7. Which of the following is not a feature of encapsulation?
    a. Data hiding
    b. Flexibility and maintainability
    c. Direct public access to attributes
    d. Modifying code without breaking other parts
8. What keyword is used in Java to create a new object?
    a. object
    b. instance
    c. new
    d. create

9. Which of the following is not a feature of Java?
    a. Platform independence
    b. Supports Object-Oriented Programming
    c. Supports multiple inheritance using classes
    d. Provides built-in security features
10. Which visibility modifier allows class members to be accessed only within the same class?
    a. public
    b. private
    c. protected
    d. default
11. Which statement is true about method overriding rules in Java?
    a. Overridden methods must have a weaker access modifier than the parent.
    b. Overridden methods must declare the same or a subclass of the return type.
    c. Overridden methods cannot throw any exceptions.
    d. Overridden methods must be marked **final**.
12. Which method type cannot be overridden in Java?
    a. public
    b. final
    c. protected
    d. default (package-private)
13. Which Java feature prevents multiple inheritance issues at the class level?
    a. Interfaces
    b. Abstract classes
    c. Static binding
    d. Packages
14. Which of the following cannot be declared abstract?
    a. Class
    b. Method
    c. Constructor
    d. Interface
15. What does the following relationship represent?

```
class Animal {}

class Dog extends Animal {}
```

    a. Dog IS-A Animal
    b. Animal IS-A Dog
    c. Dog HAS-A Animal
    d. Dog IMPLEMENTS Animal

16. Which of these keywords is used to manually throw an exception?
    a. throw
    b. throws
    c. final
    d. error
17. Which Java feature allows defining a method with the same name but different parameter lists?
    a. Overriding
    b. Overloading
    c. Polymorphism
    d. Encapsulation
18. Which of these is not true about String in Java?
    a. Strings are immutable.
    b. Strings can be stored in the String Pool.
    c. == checks for content equality.
    d. **equals()** checks for value equality.

19. Which of the following loops is guaranteed to execute at least once?
    a. for loop
    b. while loop
    c. do-while loop
    d. enhanced for loop
20. What will be the output of the following code?

```
try {

   int a = 5 / 0;

} catch (ArithmeticException e) {

   System.out.println("Divide by zero");

} finally {

   System.out.println("Finally block");

}
```

    a. Divide by zero
    b. Finally block
    c. Divide by zero followed by Finally block
    d. Runtime error
21. Which keyword is used to exit only the current loop in nested loops?
    a. return
    b. break
    c. continue

    d. exit

22. Which access modifier makes a member visible to all classes in the same package but not outside?
    a. private
    b. public
    c. protected
    d. default (no modifier)
23. What is the output of the following code?

```
for(int i = 0; i < 3; i++) {
  for(int j = 0; j < 2; j++) {
    if(i == j) break;
    System.out.print(i + "" + j + " ");
  }
```

    a. 10 20 21
    b. 01 10 20
    c. 10 21
    d. 01 21
24. What is the output of the following code snippets?

```
int i = 0;
do {
  System.out.print(i);
  i++;
} while(i < 0);
```

    a. Nothing
    b. 0
    c. Infinite loop
    d. Compiler error
25. What is the output of this nested loop?

```
for(int i = 1; i <= 2; i++) {
  for(int j = 1; j <= 2; j++) {
    System.out.print(i + "," + j + " ");
  }
```

    a. 1,1 1,2 2,1 2,2

b. 1,1 2,1 1,2 2,2
c. 1,2 2,1
d. 2,2 only

26. Which polymorphism does method overloading represent?
    a. Runtime polymorphism
    b. Compile-time polymorphism
    c. Dynamic dispatch
    d. Encapsulation
27. Which statement **best** describes abstraction in Java?
    a. It hides implementation details and exposes only essential features
    b. It allows multiple objects to share the same data.
    c. It enables one class to inherit from multiple classes.
    d. It prevents objects from being created.
28. Which of the following is **true** about Java interfaces?
    a. An interface can contain only abstract methods.
    b. A class can implement multiple interfaces.
    c. An interface can be instantiated directly.
    d. Interfaces cannot contain constants.
29. Which OOP principle is demonstrated when a `Car` class and a `Bike` class both implement a `Vehicle` interface but provide different implementations of the `start()` method?
    a. Encapsulation
    b. Inheritance
    c. Polymorphism
    d. Composition
30. Why would a developer choose an interface instead of an abstract class?
    a. To allow a class to inherit implementation from multiple classes.
    b. To support multiple inheritance of type and define a common contract.
    c. To create objects directly from the interface.
    d. To store instance variables with different values for each object.

Write the answer number for Questions 1 to 30 (a, b, c, or d) in the table below. [30 Marks]

1		**16**	
2		**17**	
3		**18**	
4		**19**	
5		**20**	
6		**21**	
7		**22**	
8		**23**	
9		**24**	
10		**25**	
11		**26**	
12		**27**	
13		**28**	
14		**29**	
15		**30**	

Part B

1. Object-Oriented Programming (OOP) emphasizes building software by modelling real-world entities. [30 Marks]
    - A. Identify two (02) differences between primitive data types and reference (non-primitive) data types in Java with examples. [04 Marks]
    - B. Briefly explain two (02) advantages of Object Oriented Programming (OOP) language. [2 x 2 = 4 Marks]
    - C. Create a class called **Patient** with variables **patientId, name, age** and **disease** with suitable data types. [03 Marks]
        - i. Implement a default constructor that assigns sample default values [02 Marks]
        - ii. Implement a parameterized constructor to initialize all variables with user-provided values. [04 Marks]
        - iii. Provide getter and setter methods for all variables, ensuring that age cannot be less than 0. [03 Marks]
        - iv. Write a method **displayPatientInfo()** that prints the patient's details in a neat format. [02 Marks]
        - v. In the main method, create two **Patient** objects: one using the default constructor and another using the parameterized constructor. [04 Marks]
        - vi. Invoke **displayPatientInfo()** for both objects. [04 Marks]

2. The strength of Object-Oriented Programming (OOP) lies in its four foundational pillars, which together form the core structure of the paradigm and define how objects and classes interact within a program. [25 Marks]
   A. Briefly explain the four pillars (concepts) of Object Oriented Programming (OOP). [4 x 2 = 8 Marks]
   B. An E-Commerce Platform consists of the following classes: **PlatformUser**, **Buyer** and **Seller.** [03 Marks]
      i. The **PlatformUser** class should have attributes like **name, email**, and **phone**, along with a method **showDetails()** that displays the basic user details. [03 Marks]
      ii. The **Buyer** class should inherit from **PlatformUser** and include additional attributes such as **buyerId** and **loyaltyTier**, as well as override the **showDetails()** method to display the buyer's details and current loyalty tier. [04 Marks]
      iii. The **Seller** class should inherit from **PlatformUser** and include **sellerId** and **storeName**, overriding the **showDetails()** method to display the seller's details and store information. [03 Marks]
      iv. In the main method, create objects of **Buyer** and **Seller**, store them in a reference of type **PlatformUser**, and call **showDetails()** to demonstrate how the correct version of the method executes at runtime [04 Marks]

3. A Java program accepts two integers from the user and performs division. However, the following situations may occur:

   - The user enters a non-numeric value.
   - The second number entered is zero.
   - The program should always display a message indicating that execution has completed.

   Answer the following questions.

   A. Identify **two exceptions** that may occur in this program and explain when each exception is thrown. **(4 Marks)**

B. Complete the following Java program by filling in the blanks with the most appropriate Java keywords, classes, or statements. **(8 Marks)**

```
import java.util.Scanner;

public class DivisionProgram {

  public static void main(String[] args) {

    Scanner sc = new Scanner(System.in);

    ________ {

      System.out.print("Enter first number: ");

      int num1 = sc.nextInt();

      System.out.print("Enter second number: ");

      int num2 = sc.nextInt();

      int result = num1 / num2;

      System.out.println("Result = " + result);

    } ________ (______________________ e) {

      System.out.println("Cannot divide by zero.");

    } ________ (______________________ e) {

      System.out.println("Invalid input.");

    } ________ {

      System.out.println("Program execution completed.");

    }

  }

}
```

C. Explain the purpose of the `finally` block in Java exception handling. **(3 Marks)**

[15 marks]

END OF THE PAPER